

Week 5: aquatic invertebrates

Individual assignment 5

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1. Set up

Packages

```
#Loading general use and cleaning packages
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

#Loading package for wrapping long axis labels
library(scales)

#Loading package for reading Excel files
library(readxl)

#Loading package for visualizing missing data
library(naniar)

#Loading package for arranging plots
library(patchwork)
```

Data

```

#Reading in taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

#Reading in aquatic invertebrate data
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects")

#Reading in site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites")

#Reading in water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling
  ↪ Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  na = c("", "n/a", "N/A", "over", "173+"))

```

Cleaning

```

#Creating a new object from site information
ncos_sites <- sites |>
  #Filtering to only include NCOS sites sampled quarterly
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")

```

```

#Creating a clean taxon list for joining with aquatic invertebrate data
taxon_list_clean <- taxon_list |>
  #Converting taxon names to snake case so they match the aquatic
  ↪ invertebrate columns
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))

```

```

#Creating a clean water quality object
water_quality_clean <- water_quality |>
  #Cleaning column names into snake case
  clean_names() |>
  #Keeping only water quality surveys from quarterly NCOS sites
  semi_join(ncos_sites, by = "site") |>
  #Creating a date-site column for joining with aquatic invertebrate data
  unite("date_site", date, site, remove = TRUE) |>
  #Grouping by date-site survey

```

```

group_by(date_site) |>
#Calculating median pH, salinity, and dissolved oxygen for each survey
summarize(med_ph = median(p_h, na.rm = TRUE),
          med_sal = median(salinity_ppt, na.rm = TRUE),
          med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
#Ungrouping the dataframe
ungroup() |>
#Filtering out the salinity outlier identified in class
filter(date_site != "2022-08-12_NVBR")

```

```

#Creating a clean aquatic invertebrate object
aq_ins_clean <- aq_ins |>
#Cleaning column names into snake case
clean_names() |>
#Keeping only sites that occur in the NCOS sites object
semi_join(ncos_sites, by = c("site")) |>
#Renaming the vial date column to date
rename(date = date_on_vial) |>
#Selecting survey information and focal aquatic invertebrate taxa
select(date, sample_type, site,
       ostracod,
       copepod,
       hemiptera_corixidae_boatman,
       cladocera,
       nematode,
       diptera_ceratopogonidae,
       annelida_oligochaete,
       diptera_chironomid,
       annelida_polychaete) |>
#Keeping only filtered beaker samples
filter(sample_type %in% c("FB 250", "FB250")) |>
#Converting taxon columns into long format
pivot_longer(cols = ostracod:annelida_polychaete,
             names_to = "taxon",
             values_to = "abundance") |>
#Grouping by date, site, and taxon
group_by(date, site, taxon) |>
#Calculating average abundance per liter from the 7.5 liter sample volume
summarize(ave_lit = sum(abundance, na.rm = TRUE) / 7.5) |>
#Ungrouping the dataframe
ungroup() |>
#Creating a date-site column for joining with water quality data

```

```

unite("date_site", date, site, remove = FALSE) |>
#Joining water quality information to each survey
left_join(water_quality_clean, by = c("date_site")) |>
#Joining taxonomic information to each taxon
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
#Adding full site names for clearer figure labels
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "South of Venoco Bridge",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelps Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NWP" ~ "West Pond",
  site == "NDC" ~ "Devereux Creek"
)) |>
#Ordering sites by median salinity so site patterns are easier to compare
mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm
  ↪   = TRUE),
  site_full = fct_rev(site_full))

```

Problems

1. Visualizing differences across sites in major

taxon abundance

a. Plan your figure

- X-Axis: `site_full`
- Y-Axis: $\log + 1$ transformed average Ostracod abundance per liter
- Geometry to represent average abundance/L: `geom_jitter()`
- Geometry to represent medians: `stat_summary()`